

Karl-Alexandre Michaud

karlthedev.tech | linkedin.com/in/karlmichaud | github.com/Karl-Michaud

EDUCATION

University of Toronto

Toronto, Ontario, CA

Honours Bachelor of Science, Computer Science (Focus in Computer Systems), Mathematics Sep 2023 – Jun 2027

- **Courses:** Software Engineering (OOP), Data Structures & Algorithms (88%), Parallel Programming (pthreads, OpenMP, MPI, Cuda), Operating Systems

EXPERIENCE

Software Developer

Sep 2025 – Present

saycamel.com

Toronto, Ontario, CA

- Built a **social platform** with **900+** users and **4.7k+** monthly visitors, connecting students to **1400+** UofT clubs and **240+** weekly events using **JavaScript (Node.js, React)**, and **Supabase database**
- Built end-to-end **ticketing and payment systems** using **Stripe API**, implementing **webhook-driven** purchase flows, **Supabase Edge Functions**, and database triggers for real-time payment event handling
- Led a team of 3, **owning** the club dashboard development and **collaborating** with cross-functional teams to design and deploy **end-to-end features**, supporting **100+** prospective clubs and **increasing engagement by 28%**

Software Developer (Markus, PythonTA)

May 2025 – Present

University of Toronto, Department of Computer Science (Prof. David Liu)

Toronto, Ontario, CA

- Engineered and maintained **MarkUs (OSS)**, a scalable web application supporting **10k+** daily users, using **Ruby on Rails, React, PostgreSQL**, and **Docker** with **CI/CD pipelines** for reliable and secure deployment
- Developed and maintained **backend services and technical documentation** for **PythonTA (OSS)**, a static code analysis tool used by **35k+** students in UofT courses since 2016, leveraging **Python** and **Astroid**
- Built **automated test frameworks** using **Pytest, Rspec, and Jest** within **CI/CD GitHub Actions pipelines**, enabling continuous integration testing for production systems
- Optimized **PythonTA** performance by **reducing import time by 9.98%** through **targeted refactoring** and **profiling** of a production codebase serving **35k+** students

PROJECTS

CExpress | *C, Shell, Make, Homebrew, Git*

karl-michaud.github.io/CExpress/

- Created **CExpress**, a lightweight, high-performance **HTTP server / microservice framework** for **C** that enables rapid prototyping and development of **RESTful APIs** with minimal overhead
- Implemented key features such as **dynamic routing, HTTP/1.1 request handling**, and low-level **TCP socket programming**, showcasing expertise in **network programming** with an **optimized memory footprint**

NorthBlue Network | *Node.js, Next.js, React, TypeScript, Supabase, Git*

www.northblue.network

- Built a scalable **networking platform** connecting UofT builders, creators, and founders, generating **10K+** impressions in 24 hours and attracting users including **4 VC investors** and notable startup founders
- Implemented auth, profile discovery, and real-time updates using **TypeScript (React, Node.js)**, and **Supabase**
- Deployed **serverless Edge Functions** via **Supabase** to handle event-driven back-end logic and real-time triggers

AI Video Generator | *C, Python, Shell, FFmpeg, Deepseek API, Whisper, Git*

www.youtube.com/watch

- Engineered an **AI video generation pipeline** in **C**, automating script generation, TTS synthesis, subtitle burning, and media merging using **FFmpeg, Whisper**, and integrating the **DeepSeek API**
- Architected a **multi-process pipeline** of 5 standalone CLI programs communicating through **Unix pipes**

Recyclobot – DeltaHacks 11 | *Swift, Python, Firebase, OpenAI API, Git*

devpost.com/software/recyclobot

- Built an AI-driven **waste management system** with **real-time object detection** and sustainability insights
- Developed back-end **microservices and data pipelines**, integrating OpenAI API and Firebase cloud database
- Delivered a functional **iOS app + hardware prototype**, ranking **top 12 overall** and **top 3 environmental**

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Python, C, Java, Ruby, Swift, HTML, CSS, SQL, Shell, Makefile

Frameworks / Libraries: React, React Native, Rails, Node.js, Next.js, SwiftUI, Maven, Pytest, JUnit, Astroid, Rspec

Developer Tools: Git, GitHub, Docker, CI/CD (GitHub Actions), PostgreSQL, Supabase